Jia Duan (Jade)

Applying for PhD positions starting Fall 2027

Nanjing University of Science and Technology · M.Eng. Candidate, Cyberspace Security
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Research Interests

World models and world foundation models; agent learning and planning; LLM safety and alignment; causal inference and counterfactual evaluation. I study reliable representation, prediction, planning, and behavioral evaluation in open, dynamic environments.

Education

- Sep. 2024 – Apr. 2027 (expected) | Nanjing University of Science and Technology | M.Eng. in Cyberspace Security
- Sep. 2019 – Jun. 2023 | Beijing Information Science and Technology University | B.Eng. in Information Security

Research Experience

Research Assistant, Center for Brain-Inspired Computing Research, Tsinghua University Jul. 2026 – present
Ongoing project: structure diagnostics for video world-model representations

- Tests whether frozen video world-model features (V-JEPA 2) encode environment factors (object position, rotation speed) as coordinate changes of shared dynamics, using structured dynamics models tested on unseen factor combinations.
- Current finding: standard held-out-trajectory validation can select spurious factor structure, while validating on held-out factor combinations removes it. Manuscript in preparation.

Manuscripts and Publications

[EMNLP 2026 Main] When Counterfactual Debiasing Backfires: Identity Cues as Task-Dependent Evidence in Harmful Language Understanding — First Author
Jia Duan, Xiaoyu Lu, Wenjing Hu, Zihao Ding

- Shows that uniform counterfactual debiasing can remove the evidence a task needs: the same identity mention is a spurious shortcut for cyberbullying detection yet label-defining evidence for target identification. Proposes task-aware representation routing, which improves target identification on three datasets while detection and sentiment stay stable.

[Under review · ARR] Identity-Edit Sensitivity and False Positives in Offensive-Language Detection — First Author

- Shows that lower identity-edit sensitivity need not mean fewer false positives: four mask-based training interventions (10 seeds each, two datasets) all reduce sensitivity, mainly by moving masked-input predictions toward the original, but none yields a statistically detectable drop in false positives on benign identity mentions; replicated on DeBERTa-v3.

[Under review · ARR] Seed and Decoding Effects on Target Attribution in Implicit Hate Explanation — First Author

- Shows that LLM explainers fine-tuned for implicit hate (15 models; Qwen2.5-7B, Llama-3.1-8B) separate targets about equally well across seeds (AUROC within 0.03) but set very different decision thresholds, so they name different target groups; also traces a causal shift to an inherited decoding default (a repetition penalty applied to prompt tokens).

[ICME 2026] Behavior-driven Hypergraph Modeling for Accurate Cyberbullying Detection in Social Media — Fifth Author

- Models social behavioral relations with hypergraphs to improve cyberbullying detection.

Patent

- Edge–Cloud Collaborative Method, Apparatus, and Computing Device for Multimodal Emotion Perception (first inventor; application accepted)

Research Projects

- Contributed to an NSFC proposal on interpretable decision rationales for multimodal cyberbullying perception.
- Contributed to multiple confidential research projects, including public-opinion analysis; details omitted.

Leadership and Activities

- China Graduate Electronics Design Contest (2025), Team Leader and Project Planning Lead: Third Prize, East China Division; First Prize, University Level.
- President, University Piano Arts Ensemble (undergraduate); docent, Harbin Military Engineering Institute Heritage Museum (graduate).